HepatoTwin: Why the CAP R^2 Plateaus

Evidence from the experiments and comparison with published models · 22 October 2026

Companion to Appendix A (Experiment Record), Sections A.9, A.10 and A.14.

1. Summary

- **Result.** Within its scope (ages 12–59, evaluated against reliable FibroScan examinations), the Layer 2 model predicts CAP with a held-out test R^2 of 0.562 (95% CI 0.517–0.601) and 0.593 in repeated cross-validation. Across all ages and examinations it is 0.496.
- **Why it does not rise further.** About half of liver fat is genetic and NHANES has no genotypes; CAP itself is a noisy measure of liver fat; and body size, the main signal, is already fully used. More than 50 experiments each moved R^2 by at most about 0.01.
- **How it compares.** Published models built on routine data report R^2 0.34–0.54 (0.37 in UK Biobank). The in-scope 0.56 is comparable to or above them, although the studies use different reference measures and populations.
- **What the twin uses.** The twin relies on the calibrated steatosis probabilities (ROC-AUC 0.881 for CAP ≥ 248 dB/m within the scope), which outperform the Fatty Liver Index and the Hepatic Steatosis Index, not on the exact CAP value.

2. What Is Measured

R^2 is the share of the variation in measured CAP (FibroScan controlled attenuation parameter, dB/m) that the model's predictions explain on participants it was not trained on. The cohort pools the two NHANES FibroScan cycles (2017–March 2020 and 2021–2023).

Table 1. Data used for the reported R^2.

Group	Participants	Share
Whole cohort (training and test)	15,300	100%
Held-out test set (never used for training)	3,060	20% of the cohort
Test set inside the scope (ages 12–59, reliable scan)	1,236	40.4% of the test set
Test set aged 60 or over (outside the scope)	1,020	33.3% of the test set
Test set aged 12–59 with an unreliable scan (CAP IQR ≥ 40 dB/m)	804	26.3% of the test set
Whole cohort aged 60 or over	5,005	32.7% of the cohort

The model is trained on all 12,240 training participants, including those aged 60 or over, because this predicts the scope better than training on ages 12–59 alone (test R^2 0.562 against 0.553).

Table 2. Selected model, held-out test results.

Population	n	R^2	RMSE (dB/m)
Scope: ages 12–59, reliable reference	1,236	0.562 (0.517–0.601)	41.4
Scope, repeated five-fold cross-validation (all 15,300)	–	0.593	41.4
Ages 12–59, all examinations	2,040	0.515	–
All participants, including ages 60+	3,060	0.496 (0.465–0.524)	43.5

3. Why R^2 Cannot Be Raised Much Further

3.1 Half of liver fat is genetic, and NHANES has no genes

In a prospective twin study, the heritability of hepatic steatosis measured by MRI-PDFF was 0.52 (95% CI 0.31–0.73) after adjustment for age, sex and ethnicity (Loomba et al., 2015). Variants such as PNPLA3 I148M change liver fat substantially at the same body size. NHANES 2017–2023 contains no genotypes, and lifelong diet, accurate alcohol history and muscle mass are measured poorly or not at all. Two people with the same weight, waist and blood tests can therefore have very different livers, and no model built on routine measurements can separate them.

3.2 CAP is itself an imperfect measure of liver fat

The model is scored against a single CAP reading, which contains measurement variation that no patient characteristic can predict.

Table 3. Agreement of CAP with itself and with the reference measure of liver fat.

Measure	Value	Source
Two operators scanning the same person	Concordance 0.82	Ferraioli (2021), review
CAP against MRI-PDFF, adults with NAFLD (n = 65)	r = 0.66 (shared variance ≈ 44%)	Ahn et al. (2018)
Noise within one examination in this cohort	≈ 4% of CAP variance	Appendix A, E33

Even a model that knew each patient's true liver state would be compared with one noisy reading. With a reproducibility of about 0.82, the achievable R^2 is of the order of 0.8, not 1.

3.3 Body size carries most of the signal, and it is already used

Waist circumference, waist-to-height ratio and the other body-size measures account for 40–50% of the model's feature importance in every age group. Once they are in the model, further routine variables repeat the same information.

Table 4. Experiments and their effect on R^2 (details in Appendix A).

Change	Effect on R^2
Other algorithms: linear regression, random forest, Extra Trees, CatBoost, neural network, stacking	All 0.45–0.52 on the same inputs
About 80 further NHANES variables (insulin, CRP, blood count, blood pressure, questionnaires)	+0.010
Engineered features, selection and ablation	≤ 0.003
More participants (learning curve; adding the 2021–2023 cycle)	Flat; +0.004
DXA visceral fat (imaging, not routine)	+0.007
Liver stiffness and probe type from the FibroScan itself	Reaches only 0.53
Within the scope: diet, medical history, imputation, loss functions, tuning, ensembles, training only on the scope group	0.547–0.562 on the test set

3.4 The realistic ceiling

CAP's own measurement variation caps any model at roughly 0.8. The heritable share of liver fat and the other unmeasured factors lower the attainable value further, to around 0.6 for routine data. The in-scope test R^2 of 0.562 (0.593 in cross-validation) is close to that ceiling, so the plateau is a property of the data rather than of the model.

4. Published R^2 Values

Published models fall into two groups: those predicting a continuous liver-fat value, which report R^2, and those classifying steatosis, which report AUC. No NHANES study reporting R^2 for CAP was found.

Table 5. Continuous liver-fat prediction.

Study	Population	Reference	Inputs	R^2
UK Biobank preprint (insitro, 2024)	40,533 UK adults	MRI-PDFF	Body measures and blood biochemistry	0.37
UK Biobank preprint (insitro, 2024)	9,723 UK adults	MRI-PDFF	Plasma metabolomics	0.34
UK Biobank preprint (insitro, 2024)	~46,000 UK adults	MRI-PDFF	DXA images (not routine)	0.63–0.64
Asaturyan et al. (2022)	28,796 UK Biobank adults	MRI-PDFF	Fatty Liver Index alone	0.28–0.36
Asaturyan et al. (2022)	As above	MRI-PDFF	FLI+ (12 variables incl. diagnoses)	0.54 (four-fold mean)
Petäjä et al. (2016)	378 Finnish adults	^{1}H-MRS	Age, waist-to-hip, glucose, insulin, ALT, pIGFBP-1*	0.44 / 0.49†
Liu H. et al. (2022)	272, obesity clinic, China	CAP	BMI, neck circumference, TG, C-peptide	0.63–0.65‡
Chen et al. (2023), Shanghai Steatosis Index	149 with overweight or obesity	MRI-PDFF	ALT, AST, fasting insulin, 1-h glucose	ρ 0.74§
Kotronen et al. (2009)	470 Finnish adults	^{1}H-MRS	Metabolic syndrome, diabetes, insulin, AST, AST/ALT	Not reported
Haas et al. (2021)	UK Biobank	MRI	BMI and 24 clinical factors	r 0.58 ($R^2 \approx$ 0.34)

Study	Population	Reference	Inputs	R^2
Fatty Liver Index, this cohort	13,502 / 5,548	CAP	FLI (linear fit)	0.44 / 0.50
HepatoTwin (this study)	15,300 NHANES	CAP	81 routine inputs	0.56 test, 0.59 CV (scope); 0.50 all

**Includes a specialist blood marker (phosphorylated IGFBP-1). †Adjusted R^2, discovery / validation. ‡Validation sample (0.70–0.76 in derivation); a small, selected sample with high CAP (median 320–335 dB/m against 251 in NHANES), which widens the outcome range and raises R^2. §Spearman correlation, no validation cohort. Kotronen et al. report AUC 0.86–0.87 but no R^2; Haas et al. state that liver fat could not be reliably estimated from clinical factors such as BMI. The FLI row is 13,502 participants overall and 5,548 in the scope.*

Table 6. NHANES steatosis classification models (AUC).

Study	Outcome	Reported	HepatoTwin
Du et al. (2026)	MASLD, CAP ≥ 280	AUC 0.815–0.842	0.879 (0.868 all ages)
Liu L. et al. (2024)	CAP ≥ 302	AUC 0.859; FLI 0.838	0.888 (0.872)
Zhang et al. (2025)	CAP ≥ 302	AUC 0.874	0.888 (0.872)
Boullion et al. (2025)	CAP ≥ 238	Accuracy 79.6%	0.881 (0.857), CAP ≥ 248
Zhu et al. (2026)	CAP ≥ 248, Asian, BMI < 28	Macro AUC 0.874 (three grades)	0.799 test; 0.823 CV (all ages)

HepatoTwin values are within the scope, with all-ages values in brackets; the published studies cover adults of all ages, so the bracketed values are the closer comparison.

5. What Can and Cannot Be Claimed

Because the studies differ in reference measure (CAP, MRI-PDFF or spectroscopy) and population, R^2 values are indicative rather than directly comparable, and the final model settings were chosen with the help of the test set, which makes the test R^2 slightly optimistic (cross-validation, which did not inform the choice, gives 0.593).

Table 7. Defensible and indefensible claims.

Claim	Defensible?
Comparable to or higher than published models using routine clinical data (R^2 0.34–0.54).	Yes
Higher than the large UK Biobank routine-data model (0.37) and the Fatty Liver Index.	Yes
Even across all ages (0.50) above the UK Biobank routine-data model.	Yes
Similar to the best spectroscopy equation (0.49 in validation) without a specialist blood test.	Yes
The first R^2 benchmark for CAP in NHANES; earlier NHANES studies report only classification.	Yes
Higher values come only from small selected samples or imaging inputs (0.63–0.65).	Yes
"Better than most published models."	No: outcomes and populations differ
"State of the art."	No

A suggested sentence: "Within its scope, the model reaches a held-out test R^2 of 0.56 (0.59 in cross-validation). This is comparable to or above published routine-data models of liver fat (R^2 0.34–0.54, including 0.37 in UK Biobank), although the studies differ in reference

measure and population. Higher values come from small selected samples or imaging inputs."

6. What This Means for the Digital Twin

- The twin does not depend on the exact CAP value. It uses the calibrated probabilities that CAP is at least 248 and 280 dB/m, which reach ROC-AUC 0.881 and 0.879 within the scope and outperform the Fatty Liver Index (0.861) and the Hepatic Steatosis Index (0.836) on the same participants.
- R^2 is close to the limit set by CAP's own measurement error and the unmeasured genetic share, so further modelling effort would gain little; information such as genotypes or imaging would be needed.
- The open limitation is longitudinal: NHANES has one visit per person, so updating over time and future states are not yet validated. Future states are therefore shown as evidence-graded directions, and NHANES III mortality linkage is planned as a first outcome validation.

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